

SPRINGBOK, South Africa

WMO: 685120

Lat: 29.6694S

Lon: 17.8789E

Elev: 3301

StdP: 13.03

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.2	39.0	5.1	8.1	65.9	9.5	10.1	65.6	35.6	50.0	31.3	50.4	9.8	100	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.1	93.5	61.0	91.2	60.5	88.8	59.8	66.5	82.4	64.9	81.8	63.5	81.3	11.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	93.0	72.6	58.9	84.1	71.1	56.8	77.7	69.6	33.1	82.6	31.8	81.8	30.6	81.1	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.5	22.9	20.3	DB	34.1	98.9	1.8	2.3	32.8	100.6	31.7	102.0	30.7	103.3	29.4	105.0	(4)
(5)			WB	31.8	68.4	1.5	2.3	30.6	70.0	29.7	71.4	28.9	72.7	27.7	74.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	72.6	73.9	72.1	67.5	61.2	55.0	54.4	54.9	59.2	63.5	65.9	69.8	(6)	
	DBStd	10.66	7.27	7.16	7.48	8.65	8.12	7.25	7.68	8.78	9.39	9.23	8.57	7.32	(7)	
	HDD50	162	0	0	0	1	8	32	47	48	19	5	2	0	(8)	
	HDD65	1769	15	9	24	78	173	304	335	333	227	144	93	34	(9)	
	CDD50	5313	702	669	687	526	355	183	182	199	294	424	478	614	(10)	
	CDD65	1444	252	257	246	152	55	5	4	19	52	99	120	183	(11)	
	CDH74	12850	2536	2511	2127	1059	237	10	10	91	383	905	1167	1814	(12)	
(13)	CDH80	4726	1038	1022	793	313	31	0	0	12	98	311	418	690	(13)	
(14)	Wind	WSAvg	9.8	9.5	9.0	8.5	9.4	9.6	10.8	11.4	10.0	10.0	10.1	10.1	9.5	(14)
(15) Precipitation	PrecAvg	7.6	0.2	0.3	0.4	0.7	0.8	1.5	1.1	1.1	0.6	0.6	0.3	0.2	(15)	
	PrecMax	14.8	1.9	2.6	2.0	3.1	3.8	4.4	3.9	3.7	2.8	1.7	2.0	2.0	(16)	
	PrecMin	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	2.6	0.4	0.5	0.4	0.7	0.9	1.1	0.8	1.0	0.6	0.5	0.5	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.5	96.8	94.1	90.8	83.6	75.3	75.4	81.6	87.7	92.3	93.9	94.1	(19)	
		MCWB	62.0	62.1	61.3	60.7	55.4	50.0	50.6	52.4	55.1	58.3	59.4	60.6	(20)	
	2%	DB	92.7	92.9	90.9	87.1	79.6	71.9	71.7	76.3	82.8	87.5	88.6	90.5	(21)	
		MCWB	61.3	61.9	60.7	59.9	54.0	49.8	50.1	51.2	53.8	56.7	57.4	59.3	(22)	
	5%	DB	89.3	90.0	87.9	83.7	76.5	69.3	68.9	72.5	78.3	83.5	85.1	87.4	(23)	
		MCWB	61.1	61.5	60.0	58.1	53.4	49.3	48.9	49.7	52.0	55.6	56.5	59.0	(24)	
	10%	DB	86.0	86.6	84.6	80.3	73.3	66.0	65.9	68.4	74.2	79.1	80.9	83.7	(25)	
		MCWB	59.9	60.6	58.8	56.9	52.2	48.0	47.8	48.2	50.9	54.4	55.6	58.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	69.1	68.1	65.7	59.7	55.4	53.8	55.5	59.2	62.9	64.0	65.1	(27)	
		MCDB	85.0	83.6	81.2	81.5	69.7	64.0	69.4	69.6	75.7	83.6	83.3	85.0	(28)	
	2%	WB	65.7	66.7	65.5	62.9	57.4	52.8	51.8	53.4	56.6	59.9	61.3	63.1	(29)	
		MCDB	83.0	82.2	80.8	79.6	70.8	64.6	67.2	70.0	74.4	79.4	79.5	82.8	(30)	
	5%	WB	63.8	65.1	63.6	60.8	55.8	51.1	50.3	51.7	54.3	58.0	59.2	61.6	(31)	
		MCDB	81.7	81.6	80.8	77.5	70.3	62.9	64.4	67.6	73.0	77.5	77.8	80.9	(32)	
	10%	WB	62.1	63.4	61.7	59.1	54.1	49.7	49.0	49.9	52.4	56.0	57.5	60.0	(33)	
		MCDB	80.6	80.9	79.8	75.4	68.8	61.1	62.3	64.3	70.2	75.2	76.0	78.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.6	21.1	20.0	17.8	15.4	14.1	14.5	16.5	19.4	21.3	22.0	22.0	(35)	
		MCDBR	23.0	21.6	20.9	19.2	17.2	17.1	16.9	19.4	22.2	23.6	24.8	23.4	(36)	
	5% WB	MCWBR	9.5	9.0	8.7	7.9	7.0	7.7	7.8	8.8	9.7	10.3	10.0	9.9	(37)	
		MCWBR	20.7	19.0	18.2	18.0	16.0	15.1	15.9	18.3	21.0	22.1	23.1	21.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.300	0.311	0.300	0.284	0.266	0.257	0.262	0.281	0.298	0.307	0.289	0.291	(40)	
		taud	2.544	2.526	2.547	2.597	2.617	2.607	2.574	2.492	2.413	2.422	2.542	2.555	(41)	
	Ebn at Noon		331	321	315	305	297	293	295	301	310	319	332	334	(42)	
		Edh at Noon	35	35	32	27	24	23	24	30	36	38	35	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2737	2490	2114	1638	1270	1095	1191	1470	1890	2332	2651	2826	(44)	
	RadStd		86	72	57	55	49	45	44	54	72	60	80	82	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)			+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page